

3(a)

**I** strong bonds

(hydrogen bonding allows) cohesion / attraction / AW, between water molecules ; AW

**A** water is cohesive      **bod** water molecules are sticky  
*in context of specific heat capacity or latent heat of vaporisation*

(provides) high specific heat capacity **and** high latent heat of vaporisation ;  
**bod** increases *for high*

(so) *idea of* resists changes in temperature ; e.g.

a lot of energy required to break H-bonds, so water remains at a (relatively) constant temperature

a lot of energy required to increase water temperature

slow to change when temperatures change

prevents short-term temperature fluctuations

**bod** acts as a temperature buffer

(so) more / lots of / AW, energy required to, change liquid water into water vapour / evaporate water ;  
*context is latent heat of vaporisation*

AVP ; e.g. example to support

| Question | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Marks    |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 3(b)     | <p>any <b>three</b> from:</p> <p><i>context is xylem wall</i><br/>           (hydrogen bonding gives) adhesion between water (molecules) and cellulose ;<br/>           (hydrogen bonding gives) adhesion between water (molecules) and hydrophilic parts of lignin ;</p> <p><i>if mp1 and mp2 not gained allow one mark for adhesion between water molecules and hydrophilic parts of xylem wall</i></p> <p><i>context is flow of water in xylem</i><br/>           adhesion, prevents column of water collapsing / helps water to move up against gravity; <b>R</b> adhesion to lignin</p> <p><i>hydrogen bonding gives cohesion between water molecules, so</i><br/>           allows a, continuous / unbroken, column (of water); <b>AW</b><br/> <b>A</b> ref. to transpiration stream <i>in correct context</i></p> <p>transpiration pull / <i>idea of</i> water moving up creates tension ;<br/> <b>A</b> water pulled up by, transpiration / evaporation / (overall) water potential gradient</p> <p><i>if no marks gained, allow one mark for ref. to cohesion-tension or cohesion <b>and</b> adhesion</i></p> | <b>3</b> |

| Question                 | Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Marks               |                    |                     |                    |                |  |   |   |                          |  |  |   |             |  |  |   |   |
|--------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------|--------------------|---------------------|--------------------|----------------|--|---|---|--------------------------|--|--|---|-------------|--|--|---|---|
| 3(c)                     | <p><i>three marks:</i><br/><b>W</b> = thymine<br/><b>X</b> = adenine ;</p> <p><b>Y</b> = cytosine<br/><b>Z</b> = guanine ;</p> <p>all spellings correct ;</p> <p><b>or</b><br/><i>two marks</i><br/>purines and pyrimidines incorrect, but still correct base pairs with correct number of H bonds ;<br/>all spellings correct ;</p> <p><b>or</b><br/><i>one mark</i><br/>purines and pyrimidines and H bonds incorrect, but still correct base pairs <b>and</b> all bases named and spelled correctly ;</p> | 3                   |                    |                     |                    |                |  |   |   |                          |  |  |   |             |  |  |   |   |
| 3(d)                     | <p><i>one mark each correct row ;</i></p> <table><tr><th>interaction</th><th>primary structure</th><th>secondary structure</th><th>tertiary structure</th></tr><tr><td>hydrogen bonds</td><td></td><td>✓</td><td>✓</td></tr><tr><td>hydrophobic interactions</td><td></td><td></td><td>✓</td></tr><tr><td>ionic bonds</td><td></td><td></td><td>✓</td></tr></table> <p><i>if no marks gained mark columns to max 2</i></p>                                                                                   | interaction         | primary structure  | secondary structure | tertiary structure | hydrogen bonds |  | ✓ | ✓ | hydrophobic interactions |  |  | ✓ | ionic bonds |  |  | ✓ | 3 |
| interaction              | primary structure                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | secondary structure | tertiary structure |                     |                    |                |  |   |   |                          |  |  |   |             |  |  |   |   |
| hydrogen bonds           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | ✓                   | ✓                  |                     |                    |                |  |   |   |                          |  |  |   |             |  |  |   |   |
| hydrophobic interactions |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     | ✓                  |                     |                    |                |  |   |   |                          |  |  |   |             |  |  |   |   |
| ionic bonds              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                     | ✓                  |                     |                    |                |  |   |   |                          |  |  |   |             |  |  |   |   |